Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 13: Approved Solutions

The Algebra of Sets

Covers Chartrand & Zhang, Section 2.2, Theorems 2.22 and 2.35

Status of this set

Ungraded practice. Solutions released with it. Attempt everything first.

Part I: How to Practice This Material

You are not learning new laws

The correspondence

Sets	Statements (Week 1)
$\cap$	$\wedge$
$\cup$	$\vee$
$\overline{}$	$\sim$
U	a tautology
$\emptyset$	a contradiction
$=$	$\equiv$

Every law in Theorem 2.22 and Theorem 2.35 is a Week 1 theorem read through this table.

Three mistakes to expect

1. **Negating the pieces but not the operation.** $\overline{A \cup B}$ is $\overline{A} \cap \overline{B}$ — the operation flips too.
2. **Treating a Venn diagram as a proof.** It shows one arrangement.
3. **Skipping the justification.** Each step of a derivation names a definition or a named theorem.

Part II: Practice Problems

Section A — Naming the source

1. For each set law, name the Week 1 result it comes from.

(a) $A \cup B = B \cup A$

(b) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

(c) $\overline{A \cap B} = \overline{A} \cup \overline{B}$

(d) $\overline{\overline{A}} = A$

Approved solution

(a) Commutative Law for $\vee$: $P \vee Q \equiv Q \vee P$ (Theorem 1.22).
(b) Distributive Law, $\vee$ over $\wedge$: $P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$ (Theorem 1.31).
This was Homework 2, Problem 3 — the one where the arithmetic analogy predicted the wrong answer, since $a + bc \neq (a + b)(a + c)$. Sets inherit the law anyway.
(c) De Morgan: $\sim(P \wedge Q) \equiv (\sim P) \vee (\sim Q)$ (Theorem 1.24).
(d) Double negation: $P \equiv \sim(\sim P)$ (Theorem 1.28).

Section B — Deriving

2. Derive $\overline{A} \cup B = \overline{A \cap \overline{B}}$, giving a reason for every step.

Approved solution

$$
\begin{aligned}
x \in \overline{A \cap \overline{B}} &\equiv \sim\big(x \in A \cap \overline{B}\big) && \text{complement}\\
&\equiv \sim\big((x \in A) \wedge (x \in \overline{B})\big) && \text{intersection}\\
&\equiv \sim\big((x \in A) \wedge \sim(x \in B)\big) && \text{complement}\\
&\equiv \big(\sim(x \in A)\big) \vee \big(\sim\sim(x \in B)\big) && \text{De Morgan (Thm 1.24)}\\
&\equiv (x \in \overline{A}) \vee (x \in B) && \text{double negation (Thm 1.28)}\\
&\equiv x \in \overline{A} \cup B && \text{union}
\end{aligned}
$$

Since membership agrees for every x, the two sets are equal.

Worth noticing what this says. By Theorem 1.48, $(\sim P) \vee Q$ is $P \Rightarrow Q$. So $\overline{A} \cup B$ is the set version of "A implies B" — and indeed $\overline{A} \cup B = U$ exactly when $A \subseteq B$.

3. Use the laws — not a membership table — to simplify $\overline{\overline{A} \cup \overline{B}}$, naming each step.

Approved solution

$$
\begin{aligned}
\overline{\overline{A} \cup \overline{B}} &= \overline{\overline{A}} \cap \overline{\overline{B}} && \text{De Morgan (Thm 2.35)}\\
&= A \cap B && \text{double complement, twice}
\end{aligned}
$$

Two steps. A membership table would also work but takes four rows and five columns to reach the same place.
This is the set version of Homework 2's challenge, where you wrote $P \wedge Q$ using only $\sim$ and $\vee$. Here it says intersection can be built from union and complement alone.

Section C — Membership tables

4. Use a membership table to determine whether $A - (B \cup C) = (A - B) \cap (A - C)$.

A	B	C	$B \cup C$	$A - (B \cup C)$	$A - B$	$(A - B) \cap (A - C)$
T	T	T	T	F	F	F
T	T	F	T	F	F	F
T	F	T	T	F	T	F
T	F	F	F	T	T	T
F	T	T	T	F	F	F
F	T	F	T	F	F	F
F	F	T	T	F	F	F
F	F	F	F	F	F	F

Approved solution

Columns five and seven agree in all eight rows, so the identity **holds**.
This is Example 2.31 in the textbook, where it is proved by double inclusion instead. Two methods, same result — and the table is faster while the element argument generalises better.

Section D — Writing

5. In five to eight complete sentences, explain why the laws for sets did not need independent proofs once the laws for statements were established. What exactly is being translated, and what is doing the mathematical work?

Approved solution

Model answer. Each set operation is defined by a logical connective applied to membership statements: $x \in A \cap B$ means $(x \in A) \wedge (x \in B)$, $x \in A \cup B$ means the disjunction, and $x \in \overline{A}$ means the negation. Two sets are equal exactly when they have the same elements, so a set identity is the claim that two membership statements are logically equivalent for every x. That means proving a set law amounts to translating both sides into statements, applying an equivalence already proved in Chapter 1, and translating back. The translation steps are dictionary lookups that use only the definitions, so the only mathematical content is the Chapter 1 equivalence in the middle. That is why De Morgan's Laws for sets follow from De Morgan's Laws for statements rather than needing a separate argument, and it is why the two lists of laws match line for line.

Full credit requires locating the mathematical work in the *logical* equivalence and the rest in the definitions. An answer that only observes "they look similar" misses the mechanism.

Section E — Challenge (optional)

6. The correspondence table pairs U with a tautology and $\emptyset$ with a contradiction. Use that to predict, without computing, what each of these equals. Then check one of them.

(a) $A \cup U$ (b) $A \cap U$ (c) $A \cup \emptyset$ (d) $A \cap \emptyset$

Approved solution

Let $\mathbf{T}$ be a tautology and $\mathbf{F}$ a contradiction. Then:
(a) $A \cup U = U$, from $P \vee \mathbf{T} \equiv \mathbf{T}$.
(b) $A \cap U = A$, from $P \wedge \mathbf{T} \equiv P$.
(c) $A \cup \emptyset = A$, from $P \vee \mathbf{F} \equiv P$.
(d) $A \cap \emptyset = \emptyset$, from $P \wedge \mathbf{F} \equiv \mathbf{F}$.
Check of (b): $x \in A \cap U$ means $x \in A$ and $x \in U$. The second is automatic, so the condition reduces to $x \in A$.

These four are the *identity* and *domination* laws. They are usually presented as extra facts to memorise; via the correspondence they are four one-line consequences of things you already know.

Instructor notes

- **Problem 1(b) is the callback worth making explicitly.** $\vee$ over $\wedge$ was Homework 2 Problem 3, the law arithmetic gets wrong. Students who fought for it then get it free now.
- **Problem 2's closing remark** — $\overline{A} \cup B$ is the set version of $A \Rightarrow B$, equal to U exactly when $A \subseteq B$ — is optional but lands well with stronger students.
- **Problem 5** is the conceptual centre. Students who can say *where the mathematics happens* understand the week; those who only notice a resemblance do not.
- **Problem 6** derives the identity and domination laws rather than listing them. Cheaper to teach this way and it reinforces the correspondence.
- Reserved for Homework 6: even-numbered Section 2.2 exercises.